Business-Intelligence Sub-System

(Sub-system #17)

Team Name: OREO
Team Number: 17
Team members:

Evan Luke D'Souza (PES1UG23AM102)	Archit Rishabh (PES1UG23AM062)	Austin D'Costa (PES1UG23AM073)

Branch & Section: AIML 'B'
Course: Object-Oriented Analysis & Design
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List of Components:

- Data Source Component
- ETL Service Component
- Data Repository Component
- Analytics Service Component
- Report Service Component
- Dashboard Service Component
- KPI Service Component
- Security Service Component
- Query Service Component

Inputs & Outputs to and from each Component:

1. Data Source Component

Interface	`IDataSource`
Inputs	ERP component data, External data files
Outputs	Structured datasets/data objects

2. ETL Service Component

Interface	`IETLService`
Inputs	raw data from `IDataSource`
Outputs	Cleaned data, transformed data, aggregated datasets

3. Data Repository Component

Interface	`IDataRepository`
Inputs	Processed data from `IETLService`
Outputs	Stored historical data

4. Analytics Service Component

Interface	`IAnalyticsService`
Inputs	data from `IDataRepository`, query parameters from the user
Outputs	Statistical results, trends, forecast results

5. Report Service Component

Interface	`IReportService`
Inputs	Analytical results, report configuration from user
Outputs	Reports (PDF, Excel), different report view models

6. Dashboard Service Component

Interface	`IDashboardService`
Inputs	KPI values, analytical results
Outputs	Chart data, widgets

7. KPI Service Component

Interface	`IKPIService`
Inputs	Analytical results, business targets
Outputs	Performance metrics, KPI status (Achieved/ Not Achieved)

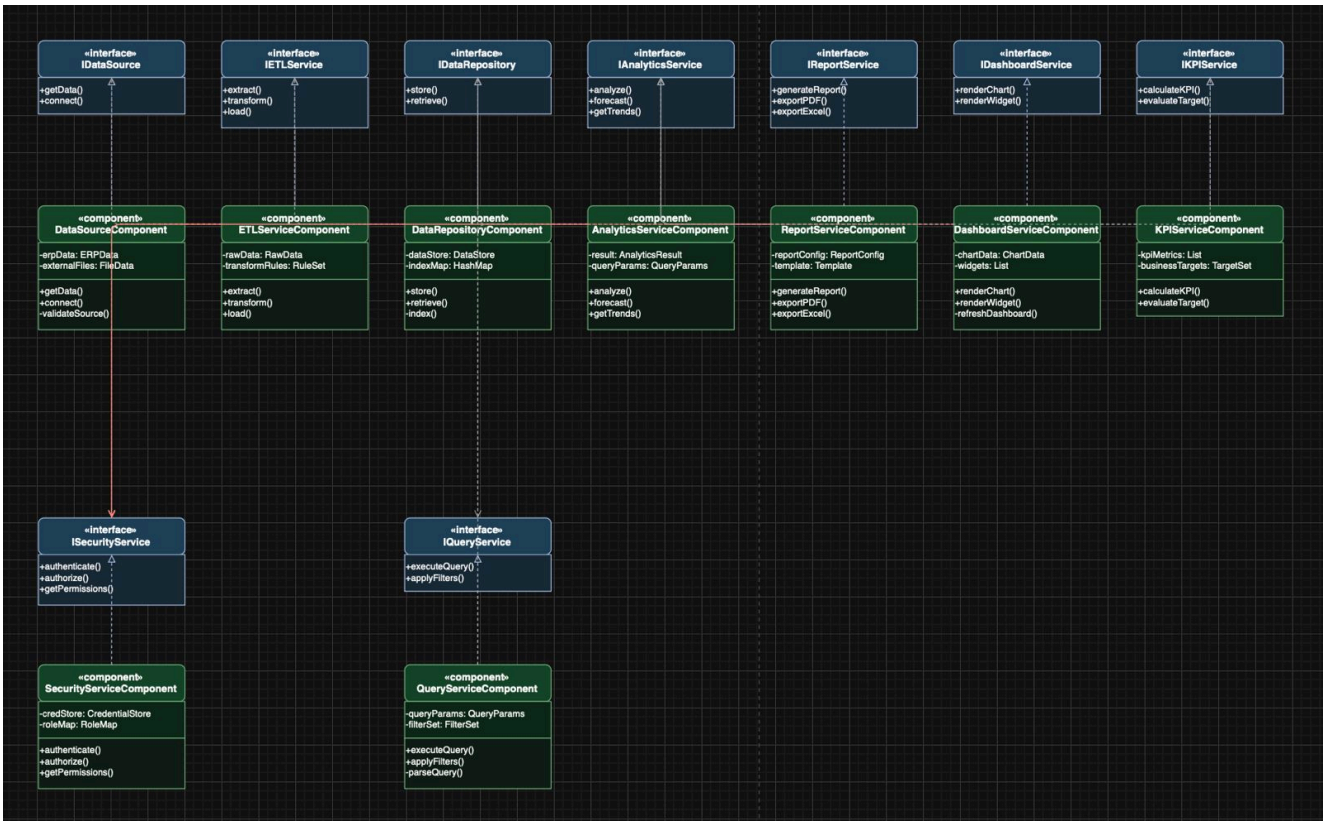
8. Security Service Component

Interface	`ISecurityService`
Inputs	User credentials, permissions and role definitions
Outputs	Access permissions, authorization decisions

9. Query Service Component

Interface	`IQueryService`
Inputs	Queries and filters specified from the user
Outputs	Filtered datasets

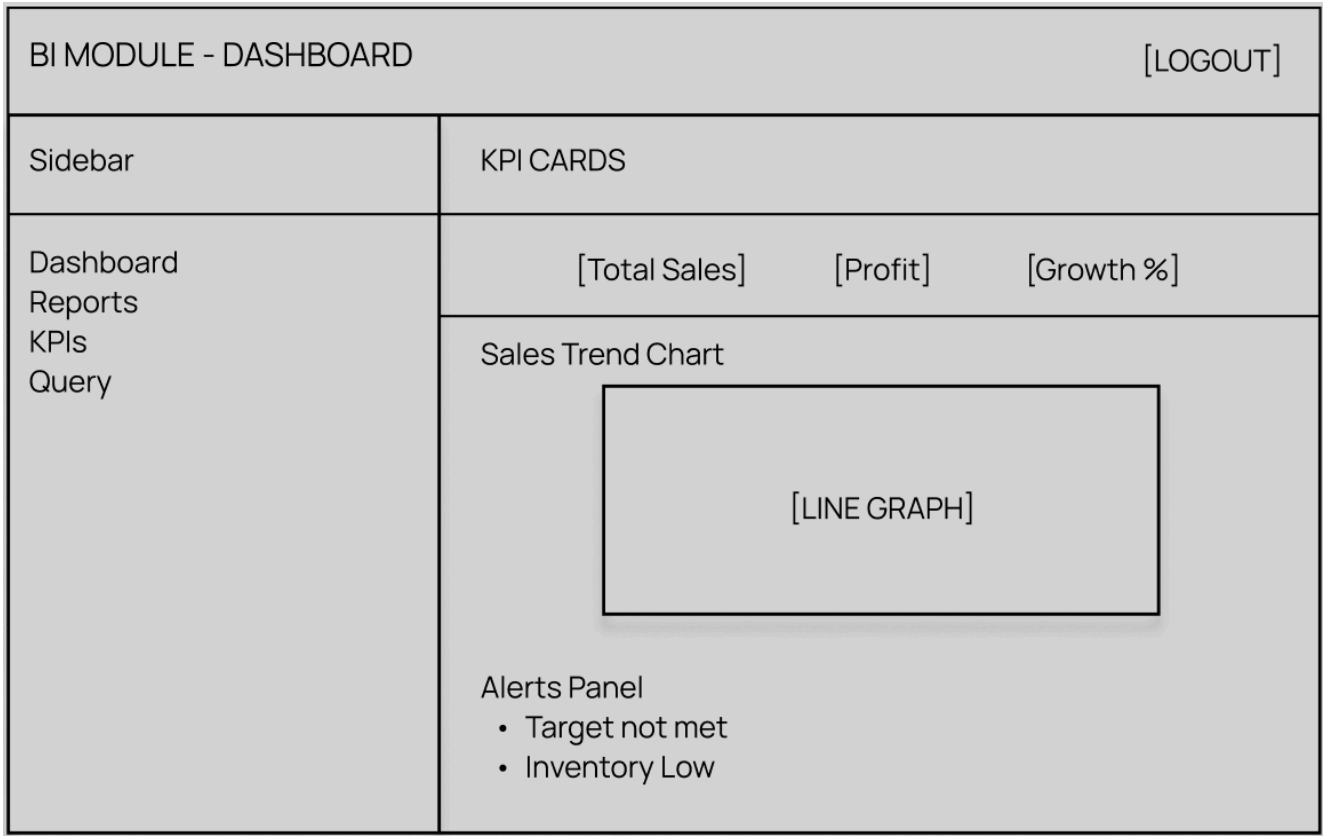
Component Diagram:



Wireframe & Minimal UI:

Login Screen:

BI Dashboard:



Report Generation:

BI MODULE - REPORTS [BACK]

Select Date Range: [From ___]
[To ___]

Select Department: [Dropdown]

Select Report Type: [Sales]

GENERATE REPORT

Report Preview Area:

[TABLE / REPORT PREVIEW]

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KPI Monitoring:

BI MODULE - KPI MONITORING [BACK]

KPI Name	Target Value	Actual Value	Status
Monthly Sales	10,00,000	8,00,000	Not Achieved
Profit Margin	20%	22%	Achieved

[REFRESH KPIs]